STAT 559 – Note 05
Inequalities and the Foundations of Lp Spaces

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1

1 Markov's inequality

For a nonnegative measurable f on any measure space and $t > 0$,

$$\mu(\{f \geq t\}) \leq \frac{1}{t} \int f \, d\mu. \tag{1}$$

For a nonnegative random variable this reads $\mathbb{P}(X \geq t) \leq \mathbb{E}X/t$.

Original TODO: Prove Markov's inequality

The pointwise inequality $t\mathbf{1}_{\{f \geq t\}} \leq f$ gives the result immediately after integration and division by $t > 0$ (reference Theorem 4.1.1).

Source page(s): 1–2

2 Jensen's and Young's inequalities

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let f be integrable with values in an interval I, and let $\phi : I \to \mathbb{R}$ be convex. When $\phi \circ f$ is integrable,

$$\phi\left(\int f \, d\mathbb{P}\right) \leq \int \phi(f) \, d\mathbb{P}, \tag{2}$$

or $\phi(\mathbb{E}X) \leq \mathbb{E}[\phi(X)]$.

Correction: Integrability in the stated form of Jensen

The handwritten theorem does not state the condition on $\phi \circ f$. The finite-integral version in reference Theorem 4.2.6 assumes it is integrable. One can also formulate an extended-value version with a well-defined right side, but the finite version suffices here.

Original TODO: Prove Jensen's inequality

Put $m = \int f \, d\mathbb{P}$, which belongs to I. If m is an included endpoint of I, nonnegativity of $f - m$ or $m - f$ and zero integral show $f = m$ a.e. Otherwise a supporting affine function exists: $\phi(x) \geq ax + b$ on I and $\phi(m) = am + b$. Integrating gives $\int \phi(f) \, d\mathbb{P} \geq am + b = \phi(m)$. See reference Section 4.2.

For conjugate exponents $p, q \in (1, \infty)$ with $1/p + 1/q = 1$, Young's inequality is

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \qquad (x, y \geq 0). \tag{3}$$

Original TODO: Prove Young's inequality

For positive x, y, concavity of the logarithm gives

$$\log\left(\frac{x^p}{p} + \frac{y^q}{q}\right) \geq \frac{1}{p}\log(x^p) + \frac{1}{q}\log(y^q) = \log(xy). \tag{4}$$

Exponentiate. If either variable is zero the inequality is immediate (reference Exercise 4.2.8). The source's phrase "natural couple" means conjugate exponents.

Source page(s): 3–5

3 Lp norms and the completeness goal

For measurable $f : \Omega \to \overline{\mathbb{R}}$ and $1 \leq p < \infty$, define

$$\|f\|_p = \left(\int |f|^p \, d\mu\right)^{1/p}. \tag{5}$$

For $p = \infty$ define the essential supremum

$$\|f\|_\infty = \inf\{K \in [0, \infty] : |f| \leq K \text{ a.e.}\}. \tag{6}$$

The finite-norm measurable functions, identified up to a.e. equality, form $L^p(\Omega, \mathcal{F}, \mu)$. The goal is to prove this is a normed vector space and is complete, hence a Banach space. A sequence is Cauchy if for every $\varepsilon > 0$ there exists N such that $\|f_n - f_m\|_p < \varepsilon$ whenever $n, m \geq N$.

Correction: A norm is defined on equivalence classes

On literal functions, $\|f\|_p = 0$ means $f = 0$ a.e., not necessarily everywhere. Thus the integral formula is only a seminorm before quotienting by a.e. equality. The quotient convention is necessary for the source's norm and Banach-space claims (reference Section 4.4).

Source page(s): 6–7

A real vector space has associative and commutative addition, a zero element, additive inverses, and scalar multiplication satisfying

$$a(bu) = (ab)u, \qquad 1u = u, \tag{7}$$
$$a(u+v) = au + av, \qquad (a+b)u = au + bu. \tag{8}$$

A norm is a finite nonnegative function on a vector space satisfying

$$\|u+v\| \le \|u\| + \|v\|, \tag{9}$$
$$\|au\| = |a|\|u\|, \tag{10}$$
$$\|u\| = 0 \iff u = 0. \tag{11}$$

Correction: Missing vector-space axiom

The source lists seven rules but omits $1u = u$. It is included above. Closure of addition and scalar multiplication is part of specifying the operations on the vector space.

The triangle inequality for L^p is Minkowski's inequality; its proof uses Hölder's inequality.

Source page(s): 8–9

4 Hölder's inequality

For conjugate exponents $1 \le p, q \le \infty$, with $1/\infty = 0$,

$$\|fg\|_1 \le \|f\|_p \|g\|_q. \tag{12}$$

At $p = q = 2$ this is Cauchy–Schwarz. For $p = 1, q = \infty$, use $|g| \le \|g\|_\infty$ a.e. to obtain

$$\int |fg|\, d\mu \le \|g\|_\infty \int |f|\, d\mu. \tag{13}$$

The opposite endpoint follows by symmetry.

Original TODO: Verify the essential-supremum bound

If $M = \|g\|_\infty < \infty$, for each n there is an a.e. bound less than $M + 1/n$. Outside the union of their exceptional null sets, $|g| \le M + 1/n$ for all n, hence $|g| \le M$. This justifies the endpoint argument.

For $1 < p, q < \infty$, first dispose of zero norms (the corresponding function is zero a.e.) and infinite right sides. For positive finite norms set $u = f/\|f\|_p$ and $v = g/\|g\|_q$. Young's inequality and integration give

$$\int |uv|\, d\mu \le \frac{1}{p}\int |u|^p\, d\mu + \frac{1}{q}\int |v|^q\, d\mu = \frac{1}{p} + \frac{1}{q} = 1. \tag{14}$$

Rescaling proves the claim (reference Theorem 4.4.1).

Correction: The normalization of the second function

The source's denominator for v resembles $\|g\|_p$. The required denominator is $\|g\|_q$, so $\int |v|^q \, d\mu = 1$. Division also requires checking that the norm is positive and finite first.

Source page(s): 10–11

5 Minkowski's inequality

For $1 \leq p \leq \infty$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p. \tag{15}$$

At $p = 1$ integrate $|f + g| \leq |f| + |g|$.

Original TODO: Verify the infinity case using null sets

Outside the union of the null sets where the two essential bounds fail,

$$|f + g| \leq |f| + |g| \leq \|f\|_\infty + \|g\|_\infty. \tag{16}$$

The definition of essential supremum gives the desired result. Ordinary pointwise suprema, suggested in the handwritten margin, do not ignore exceptional null sets and need not equal these norms.

For $1 < p < \infty$, it suffices to consider finite right side. First, convexity gives

$$|f + g|^p \leq 2^{p-1}(|f|^p + |g|^p), \tag{17}$$

so $\|f + g\|_p$ is finite. Hölder gives

$$\|f + g\|_p^p \leq \int |f||f + g|^{p-1} \, d\mu + \int |g||f + g|^{p-1} \, d\mu \tag{18}$$

$$\leq (\|f\|_p + \|g\|_p) \left\| |f + g|^{p-1} \right\|_q. \tag{19}$$

Original TODO: Verify the exponent simplification

Since $q = p/(p - 1)$,

$$\left\| |f + g|^{p-1} \right\|_q = \left(\int |f + g|^{(p-1)q} \, d\mu \right)^{1/q} = \|f + g\|_p^{p/q} = \|f + g\|_p^{p-1}. \tag{20}$$

If $\|f+g\|_p = 0$ there is nothing to prove. Otherwise divide by its finite positive $(p-1)$st power. The preliminary finiteness argument avoids an invalid cancellation of infinities (reference Theorem 4.4.3).

Source page(s): 12

6 First Borel–Cantelli lemma

If $\sum_n \mu(A_n) < \infty$, then

$$\mu(\limsup_n A_n) = 0, \qquad \limsup_n A_n = \bigcap_{m=1}^{\infty} \bigcup_{n \geq m} A_n. \tag{21}$$

Thus almost every point belongs to only finitely many A_n.

Supplement: Proof used in the completeness argument

For every m, the union bound gives $\mu(\limsup_n A_n) \leq \sum_{n \geq m} \mu(A_n)$. Tails of a convergent nonnegative series tend to zero, proving the lemma. No independence is required (reference Theorem 4.3.1).

Source page(s): 13–15

7 An almost-everywhere convergent subsequence

Lemma. Every Cauchy sequence in L^p, $1 \leq p \leq \infty$, has a subsequence converging a.e. to a measurable function $f \in L^p$. For $p < \infty$, choose increasing indices n_k with $\|f_{n_k} - f_{n_{k+1}}\|_p \leq 2^{-k}$. Define

$$A_k = \{|f_{n_k} - f_{n_{k+1}}| \geq 2^{-k/2}\}. \tag{22}$$

Markov applied to the pth power gives

$$\mu(A_k) \leq 2^{kp/2} \|f_{n_k} - f_{n_{k+1}}\|_p^p \leq 2^{-kp/2}. \tag{23}$$

The series is summable, so $N = \limsup_k A_k$ is null. Outside N the successive differences are eventually at most $2^{-k/2}$.

Correction: The good set is the complement of the limsup

Page 15 calls the infinitely-often set B, proves $\mu(B) = 0$, then writes the eventual difference bound for points in B. That bound holds on B^c. We use N for the null bad set to avoid the conflicting conventions on pages 14–15.

Source page(s): 16

Original TODO: Fill in the remaining argument; recording marker

The only writing on this page asks to "fill in" material and appears to include a recording timestamp around 35:40. The exact digits are uncertain and no recording is among the permitted sources. The mathematical continuation can nevertheless be supplied from reference Lemma 4.4.5: for $l > k$ sufficiently large and $\omega \notin N$,

$$|f_{n_l}(\omega) - f_{n_k}(\omega)| \leq \sum_{j=k}^{l-1} 2^{-j/2} \longrightarrow 0 \quad (k \to \infty). \tag{24}$$

Thus the subsequence is pointwise Cauchy outside a measurable null set and has a finite limit there; set the limit equal to zero on that set. Choose finite measurable representatives of the original L^p classes. The limit so defined is measurable, and Fatou gives

$$\int |f|^p \, d\mu \le \liminf_k \|f_{n_k}\|_p^p < \infty, \tag{25}$$

since Cauchy sequences are norm-bounded. For $p = \infty$ choose the same subsequence in essential-supremum norm, remove the countable union of exceptional sets, and use $|f_{n_{k+1}} - f_{n_k}| \le 2^{-k}$ on the remaining set. The telescoping series gives convergence and $|f| \le |f_{n_1}| + \sum_k 2^{-k}$ a.e., so $f \in L^\infty$. The completion of the Banach-space argument is supplied below.

Supplement: Completing the completeness theorem

The source stops at the subsequence lemma. To finish its stated goal, let the whole sequence be Cauchy and take the a.e. limit f of the subsequence just constructed. For $p < \infty$, if $n \ge N_\varepsilon$, Fatou applied as $k \to \infty$ gives

$$\|f_n - f\|_p^p \le \liminf_k \|f_n - f_{n_k}\|_p^p \le \varepsilon^p. \tag{26}$$

Thus the full sequence converges to f in norm. For $p = \infty$, the eventual bounds $|f_n - f_{n_k}| \le \varepsilon$ hold outside a countable union of null sets; take the pointwise subsequence limit to obtain $\|f_n - f\|_\infty \le \varepsilon$. This proves completeness for all the stated exponents (reference Theorem 4.4.6).